

PROBLEM #3

$$\frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} \frac{Lq_0}{2} \\ \frac{Lq_0}{2} \end{Bmatrix} + \begin{Bmatrix} R \\ F_1^1 \end{Bmatrix} \text{Element 1}$$

$$\frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} \frac{Lq_0}{2} \\ \frac{Lq_0}{2} \end{Bmatrix} + \begin{Bmatrix} F_1^2 \\ F_2^1 \end{Bmatrix} \text{Element 2}$$

$$\frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} F_2^2 \\ F_3^1 \end{Bmatrix} \text{Element 3}$$

$$\frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_4 \\ u_5 \end{Bmatrix} = \begin{Bmatrix} F_3^2 \\ F_4^1 \end{Bmatrix} \text{Element 4}$$

$$F_1^2 + F_1^1 = f_1$$

$$\frac{AE}{L} \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{Bmatrix} = \begin{Bmatrix} 125 \\ 250 \\ 125 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} R \\ 0 \\ 0 \\ 500 \\ 1000 \end{Bmatrix}$$

$$\frac{AE}{L} \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{Bmatrix} = \begin{Bmatrix} 125 \\ 250 \\ 125 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} R \\ 0 \\ 0 \\ 500 \\ 1000 \end{Bmatrix}$$

$$\frac{0.01 * 2 * 10^{11}}{0.25} \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} u_2 \\ u_3 \\ u_4 \\ u_5 \end{Bmatrix} = \begin{Bmatrix} 250 \\ 125 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ 500 \\ 1000 \end{Bmatrix}$$

$$\begin{Bmatrix} u_2 \\ u_3 \\ u_4 \\ u_5 \end{Bmatrix} = \begin{Bmatrix} 2.34375E-7 \\ 4.375E-7 \\ 6.25E-7 \\ 7.5E-7 \end{Bmatrix}$$

$$\text{Stress: } \sigma = \frac{F}{A}; u_x = \frac{F}{K}; K = \frac{AE}{L}$$

$$K = 8E9; A = 0.01; F = u_x * K; \text{Stress} = \sigma = \frac{u_x * 8E9}{0.01} = u_x * 8E11$$

These are NOT max Stresses. These are the stresses at the center point of each element.

$$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \end{Bmatrix} = \begin{bmatrix} 2.34375E-7 * 8E11 \\ (4.375E-7 - 2.34375E-7) * 8E11 \\ (6.25E-7 - 4.375E-7) * 8E11 \\ (7.5E-7 - 6.25E-7) * 8E11 \end{bmatrix} = \begin{bmatrix} 2.34375 * 8E4 \\ 2.03125 * 8E4 \\ 1.875 * 8E4 \\ 1.25 * 8E4 \end{bmatrix} = \begin{bmatrix} 187.5 \\ 162.5 \\ 150 \\ 100 \end{bmatrix} \text{ kPa}$$